

Foreign Demand, Production Expansion, and Deforestation: Evidence from Chinese Beef Imports from Brazil

by

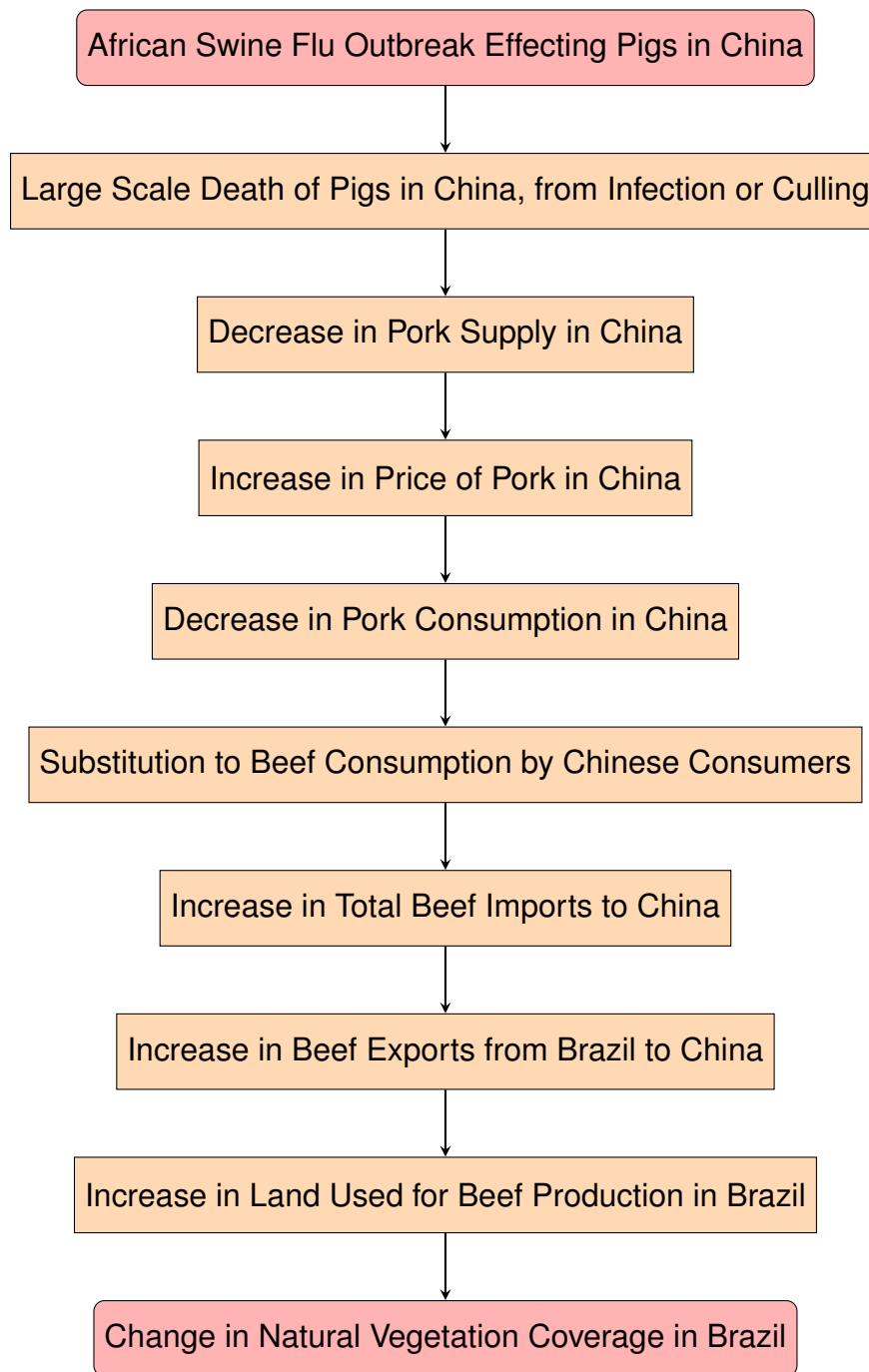
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Abstract:

This research presents an empirical analysis of the impact of a foreign demand shock on deforestation. In 2018, a disease infecting pigs in China resulted in many Chinese consumers substituting pork for beef. Much of this beef was imported from Brazil, where cattle ranching is the leading cause of deforestation. This exogenous foreign demand shock is used to evaluate the effect of foreign demand on production expansion. The study finds no evidence of a direct effect of exporting to China on forest coverage, but it cannot rule out the possibility of an indirect effect, where Chinese imports of Brazilian beef crowd out existing domestic and non-China exports, leading to production expansion and potential deforestation.

Keywords: Agriculture, Trade, Deforestation

Supervised by Clément Nedoncelle.



Flow Chart 1: Shock Transmission Mechanism

1 Introduction

For years, economists, policymakers, and the wider public have expressed worries about the environmental impacts of globalization. A key aspect of this discussion revolves around the concern that unrestricted trade promotes a relocation of production to nations where environmental regulations are lax and production methods are less efficient (Frankel and Rose, 2005; Levinson, 2009; Brunel, 2017; Abman and Lundberg, 2020).

While there is a large theoretical literature on the impact of globalization on the environment at the country level (Baghdadi et al., 2013; Brandi et al., 2020; Abman et al., 2024) and some literature focusing on emissions at a more granular level (Barrows and Ollivier, 2021; Shapiro and Walker, 2018; Gutiérrez and Teshima, 2018), there remains little evidence of the effect of trade on deforestation (Abman and Lundberg, 2020; Balboni et al., 2023; Pendrill et al., 2019).

Deforestation is important for several reasons, both local and global. Forests play a crucial role in combating climate change by acting as major carbon sinks, absorbing and storing carbon both above and below ground. When forests are cut down or damaged, significant carbon emissions are released, exacerbating the climate crisis (Gatti et al., 2021). Forests are vital for terrestrial biodiversity, housing over 80

Policies to prevent deforestation have been mostly supply-side, such as blacklisting particular municipalities and firms, implementing protected areas, monitoring, and changes in access to credit, among others. The AR6 report of the IPCC highlights the potential for demand-side policies, which could reduce emissions by 2.1 [1.1–3.6] GtCO₂-eq per year (on Climate Change, 2021). These policies could be implemented for local and foreign demand. A key issue regarding the proposed Mercosur-EU free trade agreement is the effect on the Brazilian beef industry, and subsequent effects on deforestation. The Greens/EFA Group in the European Parliament published a study, “Alternatives for a fair and sustainable

partnership between the EU and Mercosur: scenarios and guidelines,” suggesting specific rules and standards around beef imports and their origin (within Brazil) (Gehring et al., 2024). However, even if the beef imported from Brazil to Europe comes from sustainable farms, where there has been no deforestation, this increase in demand may cause indirect expansion of the industry. For example, if a farm that meets any proposed ‘anti-deforestation’ criteria that was previously selling to the domestic market (or exporting elsewhere), begins exporting to the EU, there may be expansion of the industry into forested areas to maintain supply levels to domestic markets.

The objective of this research is to evaluate the effect of foreign demand shocks on production expansion and subsequent deforestation. Foreign demand shocks can arise due to various factors such as changes in economic conditions, shifts in consumer preferences, alterations in trade policies (such as the Mercosur-EU proposed free trade agreement), or geopolitical events. To answer this research question, this paper will use municipality-level data on both beef exports and deforestation and exploit an exogenous shock to beef demand from Chinese consumers caused by the unavailability of pork due to a disease outbreak. Flow Chart 1 provides a step-by-step summary of the hypothesized cascading effects caused by the African Swine Flu outbreak, and its impact on loss of natural vegetation in Brazil.

This paper will explore each step of the proposed chain of effects, with the main interest being the last two stages, namely the expansion of beef production in Brazil and its subsequent deforestation effect. There are two hypothesized possibilities of how the expansion of beef production in Brazil may affect the loss of natural vegetation:

1. Via a ‘direct’ expansion effect, where the municipalities already engaged in exporting to China increase exports. If this expansion occurs into forested areas, there will be an increase in deforestation.

2. Via an 'indirect' expansion effect, where the municipalities already producing beef for local (or other foreign) markets that are able to begin exporting to China do so, and expansion will take place in other areas to meet local supply. If this expansion occurs into forested areas, there will be an increase in deforestation.

The study finds insufficient evidence to establish a direct, causal link between the foreign demand shock and deforestation. There currently does not exist municipality-level data for non-exported beef production in Brazil; therefore, there is insufficient data to test directly the effect of this indirect channel.

Due to a lack of granularity in the satellite data (used to measure change in forest coverage), the analysis takes place at the municipality level. Nevertheless, the model presented by Melitz, 2003, which accounts for heterogeneous firms and intra-industry reallocations, offers a potential explanation for these findings. Bernard et al., 2007 provides empirical evidence to support the model, showing that exporting firms tend to be more productive. This is not as a result of exporting, but because only the most productive firms are able to overcome the costs of entering export markets.

It is hypothesized that the most productive, existing farms operating in the local market are beginning to export to China, and new farms are in turn 'picking up the slack' to meet Brazilian (or other foreign) beef demand. Existing farms changing strategy and exporting to China will not cause deforestation, as the land is already cleared; however, the subsequent industry expansion to meet local demand (or foreign demand to countries other than China) may occur in forested areas.

1.1 Exogenous Shock to Foreign Demand: African Swine Flu

In 2018, it is estimated that 43.46 million pigs died in China, either due to ASF (African Swine Flu) virus infection, being culled to stamp out the virus or as a consequence of other ASF-related impacts during the first year of the ASF outbreaks,

accounting for 6.3% of the total number of pigs slaughtered in 2018 (You et al., 2021). This sharp decrease in local pork supply resulted in a sharp increase in the price of pork, as can be seen in Figure 1.

The pre-shock annual per capita consumption of pork was nearly 3 times greater than beef (OECD, 2024). Mao et al., 2016 note that in general, Chinese consumers consider beef a substitute to pork, although this depends on regional traditions and religion. Liang et al., 2014 finds that price and income are key factors affecting beef consumption. This explains the steady rise of Beef consumption in China before the shock, as incomes increased and then the sharp increase between 2018 and 2019, when beef suddenly became relatively cheaper (relative to pork), as seen in Figure 2 (Note that pork consumption remains far greater than Beef consumption). This increase in beef consumption was sustained even once the price of pork returned to the pre-shock levels, indicating a permanent change of preferences for some Chinese Consumers (Gale and Dong, 2023).

The increased demand for Beef resulted in an increase in Beef imports, as the Chinese Beef industry could not expand quickly enough to meet this increased demand (You et al., 2021). Figure 3 shows that pork imports also increased as a response to the local supply shock, the difference being that Pork imports then decreased (as local supply returned) however beef imports remained high.

More than half of China's beef imports come from Brazil, which in 2022 sent 55% of its total beef exports to China, more than 10% of Brazil's total beef production (WEF, 2024a). However Brazil was not always China's principal supplier of Beef. Figure 4 shows the five biggest Beef exporters to China. A steady increase in Beef Exports to China can be seen for all five countries in the periods up to and including 2018. In 2019 exports to China increase for all countries at a rate greater than the trend growth rate. Then, exports decline in 2020 in all countries except for Brazil. To measure the effect of this shock on Brazilian exports to China, the following difference-in-difference approach with two way fixed effects equation was estimated;

$$v_{it} = \beta_{1,i}(\text{Post} \times \text{Country}_i) + \gamma_i + \delta_t + \epsilon_{it} \quad (1)$$

Where v_{it} is the volume exported from country i in year t to China, $\beta_{1,i}$ is estimated for each country and is an interaction term of the shock and each country, measuring the effect of the shock specific to each country, γ_i and δ_t are the country, and year fixed effects, and ϵ_{it} is the composite error term.

Table 1: Equation 1 Estimation

	Estimate	Std. Error	t-value	P-val
$\beta_{1,Brazil}$	3.35272	0.57838	5.797	1.96e-06 ***
$\beta_{1,Argentina}$	0.93645	0.57838	1.619	0.115
$\beta_{1,Australia}$	-0.11889	0.57838	-0.206	0.838
$\beta_{1,Uruguay}$	0.01766	0.57838	0.031	0.976

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-Squared: 0.61705

Adj. R-Squared: 0.41361

F-statistic: 12.8907***

The regression estimation results shown in Table 1 show that $\beta_{1,Brazil}$ is the only significant co-variate, indicating that the effect of the shock had a significant effect only on Brazil. Brazil's Beef exports to China continue to increase rapidly each year (with the exception of 2021 which was effected by the COVID Pandemic).

There are many potential reasons the shock affected specifically Brazil. It may be that Brazil was the only country capable of rapidly increasing beef exports to China, and then once these importer-exporter links were formed, Brazil continued to expand production. Although there is no Free Trade Agreement between the two countries, they are both BRICS members, and perhaps there was some degree of preference given to Brazil Beef Exporters by Chinese Importers. This second hypothesis is less likely, given the then President, Jair Bolsonaro's reluctance to strengthen ties with China (WEF, 2024a).

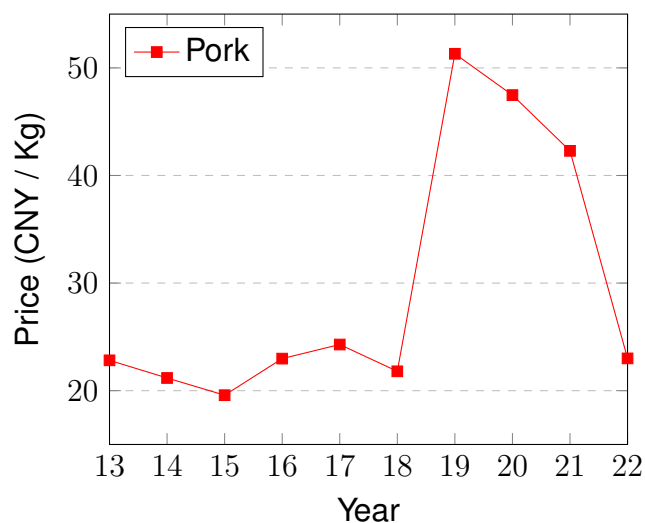


Figure 1: Price of pork, July 1st, 2013 – 2022, source: Nation Bureau of Statistitcs China, 2024

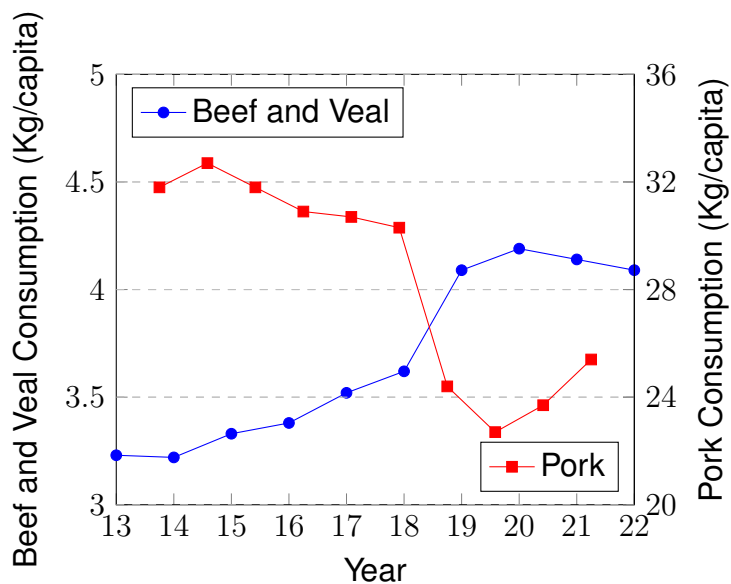


Figure 2: Chinese meat consumption, Beef and Veal vs Pork, Kilograms/capita, 2013 – 2022, source: OECD, 2024

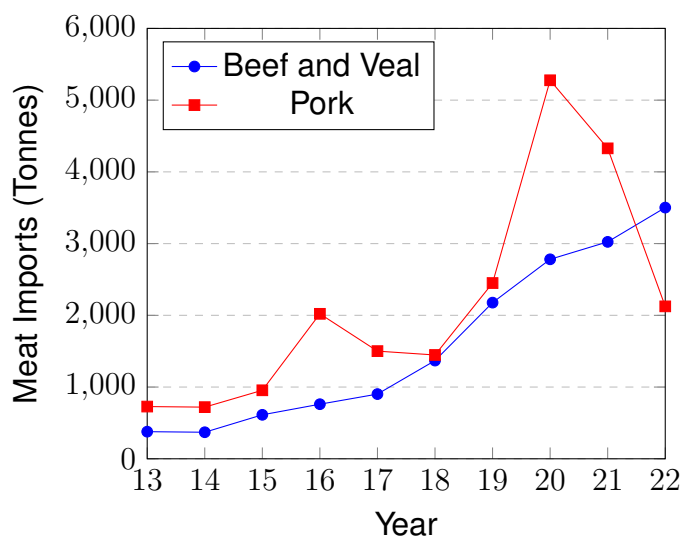


Figure 3: Chinese meat imports, Beef and Veal vs Pork, Tonnes, 2013 – 2022, source: ERS, 2024

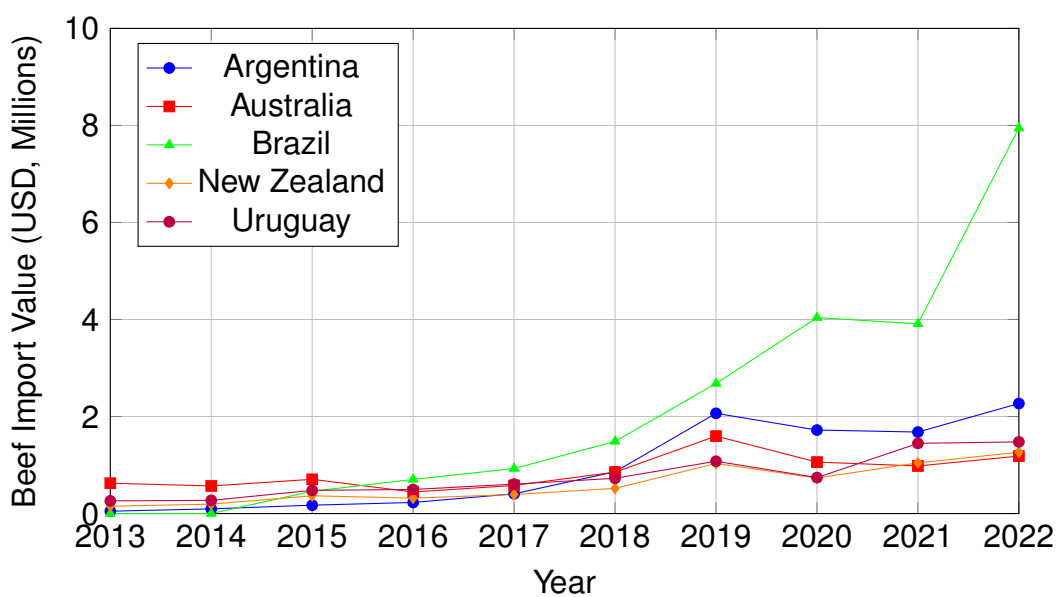


Figure 4: Chinese Beef Import Origin Over Time, source: OEC, 2024

1.2 Natural Vegetation and Agriculture in Brazil

Brazil is home to 15% to 20% of the planet's biological diversity, featuring over 120,000 invertebrate species, approximately 9,000 vertebrate species more than 4,000 plant species, and the world's largest tropical rain forest (SiBBr, 2024). It is also an agricultural powerhouse, exporting goods like soy, sugar, chicken, coffee, and beef. Brazil's agricultural sector contributes over 20% to the nation's GDP, and accounts for approximately 30% of global soybean production and 15% of the world's beef (Stabile et al., 2020). It is the second largest beef producer globally, and the largest exporter, with over 25% of Brazilian Beef being exported (Reis et al., 2023). The expansion of the agriculture industry is the leading cause of loss of natural vegetation in Brazil, resulting in the conversion of nearly 65 million hectares of native ecosystems into pastures or croplands between 1985 and 2018 (MapBiomas, 2019), with an estimated 80% of amazon deforestation being linked to cattle ranching (expansion of beef farms) (SiBBr, 2024). The contribution of this research is to disentangle the effect of foreign demand from local demand.

The majority (85%) of loss of native vegetation in Brazil has taken place in two regions, the Amazon and the Cerrado (MapBiomas, 2024). Figure 5 shows the different biomes of Brazil (geographical regions with specific climate, vegetation, and animal life).

These two biomes are used as sub samples in this paper, for two reasons. Firstly, due to their different characteristics. The Amazon is a dense tropical rain forest, and the Cerrado is made up of grasslands, Savannas and dry forests. The loss of the same amount of natural vegetation coverage (measured in area) in the Amazon is likely greater in both volume, and biodiversity, than the same area in the Cerrado. Because of these different characteristics, loss of natural vegetation in the Amazon is considered deforestation, whilst most loss of natural vegetation in the Cerrado is not (WEF, 2024b). Secondly, The pace of clearing in the Cerrado has increased over the last two decades, whilst trends in deforestation in the Amazon have however slowed Strassburg et al., 2017. Continued clearing

of the Cerrado is likely, as it has relatively little legal protection compared to the Amazon biome, which is subjected to various conservation policies Strassburg et al., 2017. Between 2004 and 2017, annual deforestation rates in the Amazon decreased by more than 70% (INPE, 2018a, INPE, 2018b), while soybean and beef production increased by 130% and 72%, respectively (IBGE, 2019, IBGE, 2018). These reductions in deforestation have been attributed to various public and private policies (Alix-Garcia et al., 2017, Boucher et al., 2013). This decrease in deforestation rates, and increase in agricultural production indicate that it is possible to increase production without causing deforestation. Several studies suggest that Brazil now has an opportunity to significantly increase its agricultural production without further deforestation (Chambers and Artaxo, 2017, Garcia et al., 2017). This potential stems from the fact that Brazil has a vast amount of already cleared, underutilized land and a capable, innovative agricultural sector (Strassburg et al., 2014).



Figure 5: Biomes of Brazil

2 Literature Review

This research builds upon and contributes to three strands of the literature. Firstly, it adds to the growing body of research on the link between trade and deforestation. Previous studies have highlighted several mechanisms through which trade openness may influence deforestation, including changes in agricultural market prices (Berman et al., 2023), land-use value, costs of imported agricultural inputs, and productivity (Abman and Lundberg, 2020). Despite these insights, empirical evidence on the impact of trade on deforestation has remained sparse (Balboni et al., 2023). Recent research, however, has shed new light on this relationship. Abman and Lundberg, 2020 demonstrates that deforestation tends to rise following the implementation of RTAs, with agricultural trade serving as a key mediator. Furthermore, Pendrill et al., 2019 reveals that a significant portion of tropical deforestation is driven by consumer demand in foreign countries, particularly for soybeans. This research will consider a particular case where foreign demand to one particular destination increased sharply.

Secondly, this research contributes to the growing body of literature focused on policies designed to mitigate deforestation. Recent research has highlighted the potential of trade policies to encourage conservation efforts in countries at risk of deforestation (Balboni et al., 2023). For instance, some studies have proposed that import tariffs could be an effective measure to curb deforestation (Harstad, 2024; Dominguez-lino, 2021), while others have examined the impact of bans (Busch et al., 2022; Villoria et al., 2022). This shows that attempts to regulate farms that are exporting may be missing the problem as the effect of deforestation may be indirect.

Thirdly, the research is related more broadly to the literature on agriculture and deforestation in Brazil. Several recent papers focus on soybean exports and their effect on deforestation (Crepin and Nedoncelle, 2023; Villoria et al., 2022; de LT Oliveira, 2017). There are also studies on the effect of the beef industry (Nepstad et al., 2014; Skidmore et al., 2021) with some focusing particularly on beef ex-

ports (Zu Ermgassen et al., 2020; Schierhorn et al., 2016). However, none have exploited a random experiment approach using a specific destination market, as is done in this research.

The rest of the paper is organized as follows. The next section details the data sources and presents the main variables used in the empirical analysis, as well as presenting some descriptive data. Section 4 provides an overview of the empirical methodologies employed and describes the main results of the analysis. The final section concludes and discusses the results.

3 Data

The data used covers beef export quantity and origin, reporting the municipality of production, and destination of exports, as well as satellite imagery of forest coverage at the municipality level.

3.1 Beef Exports

To identify beef export flows, the TRASE database (zu Ermgassen et al., 2023) is used. This database is assembled from customs records and delineates trade flows in terms of physical quantities (referred to as volume) and port of export FOB values, spanning from source municipalities to destination countries, along with data on land used by each exporter. The data includes 3746 municipalities where beef was produced for exportation, from 2010 to 2020 (with no data for 2018). The database provides individual data points for each exporting farm in the municipality; however, there is no farm-level identifier to track the change in land use and/or export destination over time for a specific farm. Therefore, these farms are aggregated within municipalities for each year, based on the destination of the exports, resulting in an aggregate measure of volume and land use for exports from each municipality to each country, for each time period.

3.2 Deforestation

To measure deforestation, the MapBiomass database (Souza Jr et al., 2022) is used, aggregated at the municipality-year level. This database measures natural vegetation coverage throughout Brazil from 1985 to 2022, and contains data on the yearly change in coverage for each municipality. This research considers the MapBiomass classifications of forest, savanna, flooded forest, wetland, and grassland in the analysis of 'forest coverage'.

3.3 Descriptive Statistics

3.3.1 Beef Exports and Land Use Increasing

Figure 6 shows that there has been an increase in the volume of beef produced for export in each biome, particularly in the Cerrado, where roughly half of all Brazilian beef that is exported is produced. It is important to note where the industry expansion is occurring, as loss of vegetation in the Amazon is considered true deforestation, whereas the Cerrado is an ecosystem of tropical savanna not included in the forest definition. However, vegetation loss in this region is still important (WEF, 2024b). The Mata Atlantica biome is considered forest but is largely already deforested (WEF, 2024b). Figure 7 shows the total land used for beef produced for export, and reflects the results from Figure 6, indicating that land used to farm beef has increased in the Amazonia and Cerrado biomes.

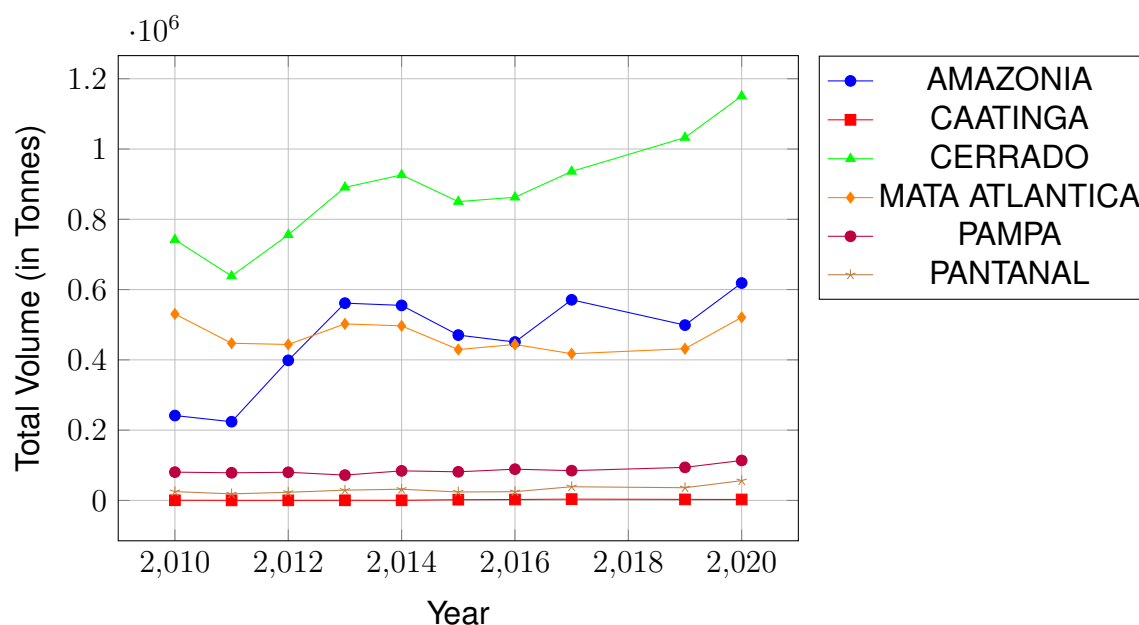


Figure 6: Total Volume of Exported Beef by Biome

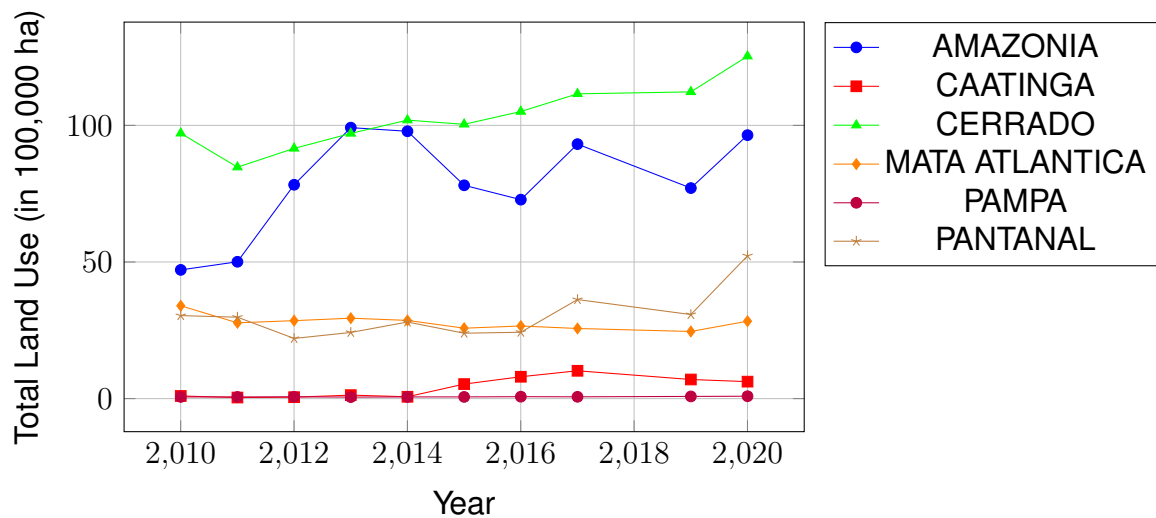


Figure 7: Total Land Use to Produce Exported Beef by Biome

3.3.2 Exports to China Potentially Replacing Exports to Rest of World

Figure 8 shows that while Chinese imports of beef from Brazil have been increasing steadily, the 'Rest of World' imports of beef from Brazil had remained relatively constant. The picture post-shock is less clear. In the first period after the shock, there was an increase in volume exported to both China and all other countries. However, between 2019 and 2020, there was a decrease which may indicate some 'crowding out' effect by China in the market for Brazilian beef, whereby increased exports to China are coming at the cost of beef exports to other countries. Figure 9 shows a marked decrease in land used to cultivate beef that is exported to countries other than China after the shock in 2018. This is further evidence that the increase in exports to China may have replaced exports to other countries. Although there is no municipality-level data on local beef production, data from the Brazilian Institute of Geography and Statistics show that local beef consumption in Brazil has remained stable over the last 20 years (IBGE, 2023). This makes it unlikely that an increase in Brazilian beef exports to China has replaced local beef consumption.

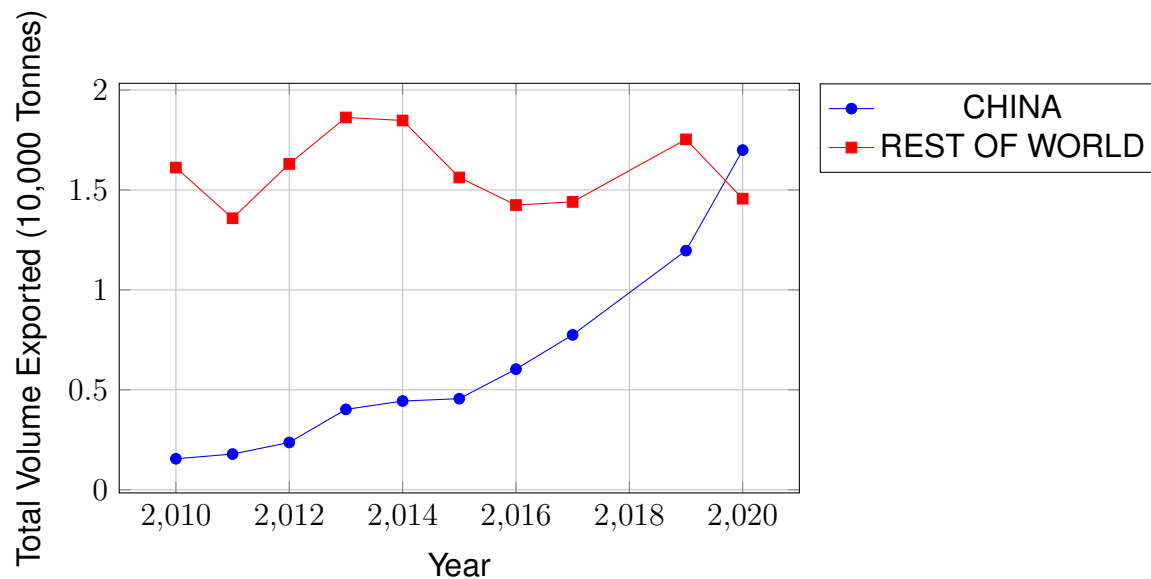


Figure 8: Total Volume of Exported Beef by Destination (CHINA vs. REST OF WORLD)

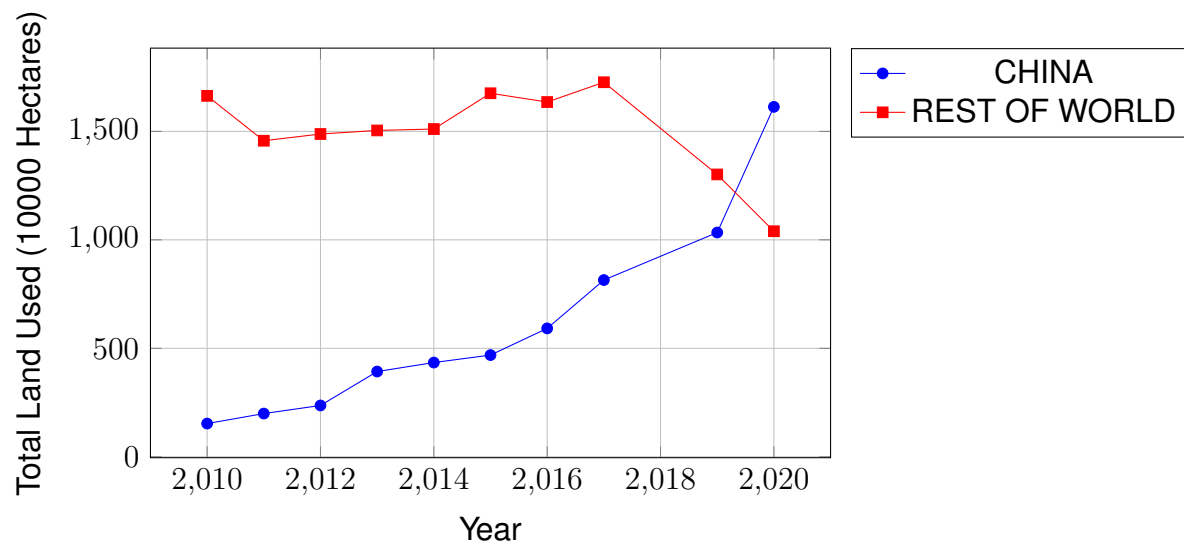


Figure 9: Total Land Use to Produce Beef for Exports by Destination (CHINA and REST OF WORLD)

3.3.3 Continued Loss of Forest Coverage

The MapBiomass data shows clearly that there has been a continued loss of forest cover in all biomes (except for some small increases in the Mata Atlantica). Two of the regions that consistently see negative forest change are the two regions that produce the most beef for export, The Amazon and The Cerrado.

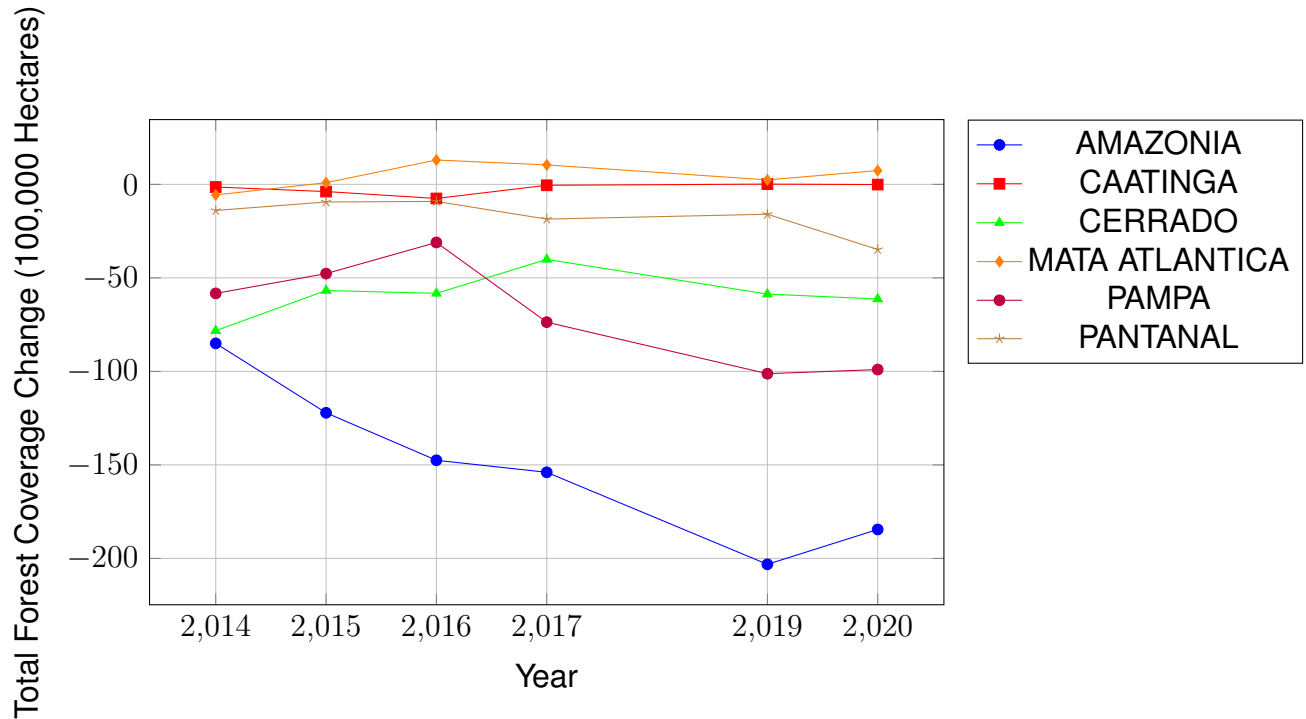


Figure 10: Total Yearly Forest Cover Change by Biome

4 Empirical Strategy & Results

4.1 Exported Volume and Land Use

Equation 2 shows the regression used to estimate the effect of change in volume Δv_{it} exported by municipality i in year t , on change in land coverage Δl_{it} in municipality i in year t . It is estimated using two-way fixed effects, controlling for municipality μ_i and year τ_t fixed effects. ϵ_{it} is the error term. The regression does not consider destination country-specific effects, nor the shock.

$$\Delta l_{it} = \beta_1 \Delta v_{it} + \mu_i + \tau_t + \epsilon_{it} \quad (2)$$

Table 2: Estimation of Equation 2

biome	All	Amazonia	Cerrado
β_1	12.624***	17.049 ***	11.331 ***
Observations	19290	1801	5538
R-squared	0.726	0.729	0.884
F-statistic	31393***	3001.4***	18986*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*'

Table 2 shows the estimates for Equation 2. The β_1 estimate is significant and positive for the total sample, as well as the Amazon and Cerrado samples. Considering the all-biomes sample, the regression estimates that a 1 tonne increase in exported volume causes a 12.624 hectare increase in land use. The effect is largest for the Amazon subset, indicating that for the same increase in volume exported, more land is used to produce in the Amazon.

Equation 3 considers the impact of the shock. As in Equation 2, β_1 once again measures the effect of Δv_{it} on the dependent variable Δl_{it} . Additionally, dummy variables 'China', and 'Post2018' are included. The dummy variable 'China' is

equal to 1 when the export destination is 'China', and 0 otherwise, and the dummy variable 'Post2018' is equal to 1 when the year is after the shock (2018), and 0 otherwise. Therefore, β_2 is an estimate of the effect of exporting to China (before or after the shock) and β_3 is the coefficient of the interaction term of the two dummy variables. The interaction term will be equal to one if and only if the year of the observation is after 2018, and the destination of the export is China. μ_i and τ_t are the municipality and year fixed effects, and ϵ_{it} is the composite error term.

$$\Delta l_{it} = \beta_1 \Delta v_{it} + \beta_2 \text{China}_{it} + \beta_3 (\text{China}_{it} \times \text{Post2018}_t) + \mu_i + \tau_t + \epsilon_{it} \quad (3)$$

Table 3: Estimation of Equation 3

Biome	All	Amazonia	Cerrado
β_1	12.686***	17.082 ***	11.365 ***
β_2	-54.803	34.590	27.465
β_3	-309.43***	-343.26**	-262.367 ***
Observations	307358	31354	113941
R-squared	0.729	0.732	0.886
F-statistic	272853***	28187.9***	292452***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*'

Table 3 shows the estimates for Equation 3. The β_1 estimate is significant and positive for the total sample, as well as the Amazon and Cerrado samples. β_2 is not significant, indicating that exporting to China had no effect on the overall relationship between volume change and land use change, measured by β_1 . β_3 shows interesting results. The estimate is negative and significant for each sample, and indicates that exports to China after the shock required less land to produce the same amount of volume. Producing the same volume of beef with less land use implies greater productivity. This implies that the exports to China after the shock came from municipalities where productivity was higher. This is consistent with the Melitz, 2003 model and the empirical findings of Bernard et al., 2007. These

relatively more productive firms exporting to China likely mean that beef industry expansion happens elsewhere to maintain production levels for local consumption. If this expansion occurs in forested areas, this would be evidence of an 'indirect effect'; however, there is currently insufficient data to test this hypothesis.

4.2 Land Use and Forest Coverage

This subsection will analyze the effect of increased land use on forest coverage. The first regression, Equation 4, is used to estimate the effect of change in land use Δl_{it} exported by municipality i in year t , on change in forest coverage Δf_{it} in municipality i in year t . It is estimated using two-way fixed effects, controlling for municipality μ_i and year τ_t fixed effects. This regression does not consider the destination country of exports, nor any shock.

$$\Delta f_{it} = \beta_1 \Delta l_{it} + \mu_i + \tau_t + \epsilon_{it} \quad (4)$$

Table 4: Estimation of Equation 4

Biome	All	Amazonia	Cerrado
β_1	-0.008***	-0.008***	-0.001 ***
Observations	19290	1801	5538
R-squared	0.002	0.001	0.0009
F-statistic	30.617***	2.966	0.571

Signif. codes: 0 '***' 0.001 '**' 0.01 '*'

Table 4 shows that β_1 estimates from Equation 1 are significant and with the expected signs for the whole sample, as well as biome subsets. When considering the point estimate for the 'all' biomes estimate, it can be understood as a one-hectare increase in land use resulting in a decrease of 0.008 hectares of forest coverage. The estimated effect is larger in magnitude for the Amazon than the

Cerrado, and this may reflect the differences in density of natural vegetation coverage between the Amazon and Cerrado biomes. The R-squared values are low for each sample, and the F-test is only rejected for the 'all' biomes sample, as the regression suffers heavily from omitted variable bias that is not accounted for by the fixed effects. The most obvious omission is land use by other agricultural industries, notably by non-export beef production, which accounts for approximately 75% of total beef production (Reis et al., 2023); however, there is currently no data on the location of production for domestically consumed beef.

Equation 5 considers the impact of the shock. As in Equation 4, β_1 once again measures the effect of Δl_{it} on the dependent variable Δf_{it} . Additionally, dummy variables 'China', and 'Post2018' are included. The dummy variable 'China' is equal to 1 when the export destination is 'China', and 0 otherwise, and the dummy variable 'Post2018' is equal to 1 when the year is after the shock (2018), and 0 otherwise. Therefore, β_2 is an estimate of the effect of exporting to China (before or after the shock) and β_3 is the coefficient of the interaction term of the two dummy variables. The interaction term will be equal to one if and only if the year of the observation is after 2018, and the destination of the export is China. μ_i and τ_t are the municipality and year fixed effects, and ϵ_{it} is the composite error term.

$$\Delta f_{it} = \beta_1 \Delta l_{it} + \beta_2 \text{China}_{it} + \beta_3 (\text{China}_{it} \times \text{Post2018}_t) + \mu_i + \tau_t + \epsilon_{it} \quad (5)$$

Table 5: Estimation of Equation 5

Biome	All	Amazonia	Cerrado
β_1	-0.006***	-0.009.	-0.001
β_2	-48.45*	-89.17	-25.25
β_3	148.64**	253.822	43.79
Observations	307358	31354	113941
R-squared	0.0002	0.0001	0
F-statistic	17.702***	1.546	1.598

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.'

The null hypotheses for the F-tests for equations estimated using the Amazon and Cerrado sub-samples cannot be rejected, indicating that there is no significant effect of any of the variables in these two regressions. The β_3 estimate is positive and significant for the full sample, indicating that the shock to Chinese demand for beef caused an a relative positive impact on forest coverage in municipalities that exported to China, compared to those municipalities that did not. This does not mean that municipalities exporting to China after the shock experienced a positive change in forest coverage, as the net impact on forest coverage depends also on the β_2 , and β_1 impacts. The β_2 estimate implies land used to produce beef that was exported to China decreased forest coverage by 48.5 hectares more than farms that exported to other countries. β_1 is consistent with equation 4, and implies that a one-hectare increase in land use for beef exports results in a decrease of 0.006 hectares of forest coverage.

These results are likely unreliable. These sub-sample areas are the largest, most deforested, and highest exporting biomes, and the effect of the shock is not significant. As with estimates for Equation 4, there is likely an omitted variable bias issue. The proportion of land (on average) within each municipality that is used to produce beef that is exported to China is likely very small, and there are many other variables that would affect forest coverage, some of which are likely correlated with the production of beef exports to China, causing a bias in the model.

5 Conclusion

The objective of this research is to evaluate the effect of foreign demand shocks on production expansion and subsequent deforestation. Using a disease outbreak affecting pork in China as an exogenous shock, the paper explains the mechanisms of how this ultimately impacted beef production patterns in Brazil. Two possible impact channels were hypothesized.

The first hypothesis of a direct expansion effect caused by the foreign demand shock is not supported. The econometric analysis shows that, while overall land use for beef production that is exported has a negative impact on vegetation coverage, the impact of a specific foreign demand shock is unclear, at least at the municipality level.

The second hypothesis of an indirect expansion effect is inconclusive. The finding that the increase in volume exported to China was fulfilled by more productive producers, coupled with the data from IBGE, 2023 that shows domestic consumption of beef in Brazil has remained constant, indicates that there is likely some expansion in production in the domestic beef market to maintain local supply. This expansion would be from less productive firms, which may have an effect on forest coverage. There is currently no data on the location of producers selling in the local market, however, a database, 'Do Pasto ao Prato' (From Pasture to Plate), is currently being developed to provide this data that could be used in future research to test this hypothesis directly.

These inconclusive results, however, have some potential policy implications. In the context of the Mercosur-EU trade deal, the Gehring et al., 2024 report discusses import origin deforestation standards. This research shows that an increase in exports as a result of foreign demand shock (in this example, tariff reductions) will lead to the relatively more productive firms exporting to the EU and indirect expansion occurring elsewhere to maintain supply to the domestic market or other foreign markets, which may cause deforestation. Any policies to

attempt to minimise potential deforestation impacts of a trade deal would need to be industry-wide.

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